

Prashant Bhattarai

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Education

Mississippi State University

B.S. in Mechanical Engineering; GPA: 4.00/4.00

- Judy and Bobby Shackouls Honors College; President's List (all semesters)

Starkville, MS

Expected May 2028

Experience

Undergraduate Researcher

Jan 2026 – Present

Center for Advanced Vehicular Systems (CAVS), Mississippi State University

Starkville, MS

- Leading a UAV-assisted map recovery project for autonomous ground vehicles; deployed ROS2 Nav2 with DLIO-based LiDAR-IMU SLAM on a Jetson-powered RC platform, achieving an 80% navigation recovery rate across 40 simulation trials
- Developing a perception-based leader-follower multi-vehicle system fusing camera, LiDAR, and IMU data, progressing from ArUco tracking to ML-based vehicle detection
- Designed and 3D-printed sensor/compute mounts for RC research platforms (Jetson, LiDAR, IMU, camera) using SolidWorks and DFM principles
- Benchmarked decentralized multi-agent navigation architectures against a centralized M*-based Conflict-Based Search (CBS) algorithm in ROS2/MAVS simulation for path efficiency and inter-agent latency

Undergraduate Research Assistant

Aug 2025 – Nov 2025

Industrial & Systems Engineering, Mississippi State University

Starkville, MS

- Designed and fabricated structural components and lattice structures via FDM, resin, and metal AM for ceramic nanocomposite research improving fracture toughness (Buried Combustion Method)

Product Design Research Intern

Jun 2025 – Aug 2025

USDA REEU – AI2F Summer Research Program, Mississippi State University

Starkville, MS

- Led end-to-end mechanical design and prototyping of a portable AI-enabled device, creating CAD models and assemblies across multiple hardware iterations

Undergraduate Research Assistant

Feb 2025 – Apr 2025

ISTVS Student Chapter, Mississippi State University

Starkville, MS

- Investigated and benchmarked GPS-only vs. GPS/IMU sensor fusion localization on a Polaris MRZR in off-road environments; collected, cleaned, and analyzed positioning data

Projects

Computer Vision & Robotics Perception | C++, OpenCV

- Developed a C++ spatial perception pipeline for AMR docking (Fiducial Docking System), using OpenCV's solvePnP on multi-marker ArUco grids for robust real-time 6-DoF pose estimation that stays locked even under heavy occlusion, fused with a custom 6-state Kalman filter to eliminate 3D tracking jitter
- Built a vision-based industrial safety interlock (Intrusion Detection) using MOG2 background subtraction, morphological noise filtering, and image-moment centroid tracking, cutting processed pixel area over 60% while resisting lighting-based false alarms

Synthetic Data Generation & Sim-to-Real Object Detection | NVIDIA Isaac Sim, YOLO11, Python, ROS2

Ongoing

- Built a synthetic-data generation pipeline in NVIDIA Isaac Sim Replicator and trained a YOLO11s detector for fallen cardboard boxes (0.99 validation mAP50-95); diagnosed a sim-to-real generalization gap from testing against real photos and am iterating on the dataset while building a ROS2 node for real-time inference

Non-Linear Model Predictive Control (MPC) for Cart-Pole System | Python, NumPy, CasADi

Ongoing

- Derived non-linear state-space dynamics for an inverted pendulum from first-principles Newtonian mechanics and built a custom RK4 physics simulation engine in Python for high-fidelity state prediction
- Formulated a discrete-time non-linear MPC controller in CasADi to solve constrained real-time optimization, enforcing state/input limits (motor torque, track length) for eventual hardware deployment

Technical Skills

Autonomy & Robotics: ROS2, Nav2 Path Planning, LiDAR-IMU SLAM (DLIO), Sensor Fusion (Camera/LiDAR/IMU), Multi-Agent Path Planning (M*, CBS), Perception-Based Tracking (ArUco, ML-Based Detection), MAVS Simulation

CAD & Systems Integration: SolidWorks (CSWP), Fusion 360, GD&T, Engineering Drawings, Sensor/Compute Packaging (Jetson, LiDAR, IMU, Camera)

Manufacturing & Analysis: 3D Printing (FDM/Resin/Metal), CNC/CAM, Sheet Metal, FEA, DFM, Rapid Prototyping

Programming & Systems: Python, C, C++, Linux, Windows

Leadership & Activities

Co-Lead, Vehicle Hardware Integration – MSU EcoCAR Innovation Challenge Team

Aug 2026 – Present

- Co-lead the Vehicle Hardware Integration subteam, physically integrating vehicle systems and components (sensors, wiring, mounts, electronics) into the EcoCAR platform in Siemens NX; building a GitLab-based version control workflow for the team's shared CAD files, since NX's native version control isn't compatible with the university's HPC setup

Chief of Additive Manufacturing – Advanced Manufacturing Society

Feb 2025 – Present

- Lead hands-on workshops in additive manufacturing, 3- and 5-axis CNC machining, milling, lathe work, FEA, and CAD; hold CSWA (Mechanical Design), CSWA (Additive Manufacturing), and CSWP certifications

Vice President – International Society for Terrain-Vehicle Systems (ISTVS)

Apr 2025 – Present

- Lead membership and research engagement initiatives; collaborating with Virginia Tech and the University of Pretoria on the society's first-ever student competition in wheel and tracked vehicle design – designing a wheel in CAD and simulating it in YADE DEM to optimize the result